

MODULE 2

Data warehouse Implementation

1.1 Efficient Data cube computation

Data warehouse contains huge volume of data. OLAP servers demand that decision support queries be answered in the order of seconds.

Therefore, it is crucial for data warehouse system to support highly efficient data cube computation techniques, access methods and query processing techniques.

Multidimensional data analysis is the efficient computation of aggregation across many sets of dimensions

In SQL, aggregation is referred to as group-by's where Each group-by is represented by a cuboid. Set of group-by forms a lattice of cuboid

Compute cube operator

→ **Compute cube operator:** Computes aggregates over all the subsets of dimension

→ This requires excessive storage space, especially for large no of dimensions

Example:

Suppose that you want to create a data cube for AllElectronics sales that contains:

city, item, year and sales_in_dollar.

You want to be able to analyse the data with the following queries:

“compute the sum of sales, grouping by city and item”

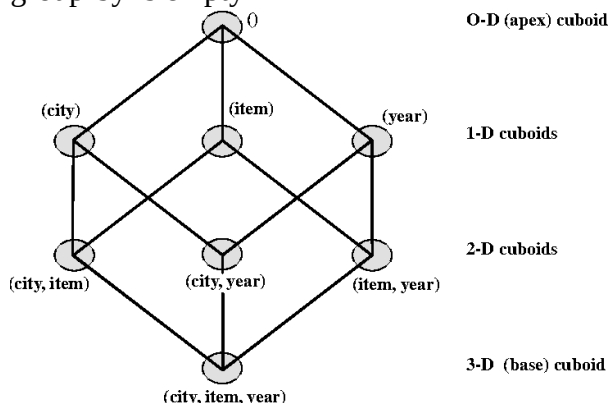
“compute the sum of sales, grouping by city”

“compute the sum of sales, grouping by item”

The total no of cuboid, or group-by's that can be computed for this data cube is $2^3=8$

The possible group-by's are as follows:

{(city, item, year), (city, item), (city, year), (item, year), (city), (item), (year), ()} where () means group-by is empty



→ The base cuboid contains all three dimensions city, item and year.

→ The apex cuboid or 0-D cuboid refers to the case where the group by is empty. It contains the total sum of all the sales

→ the base cuboid is the least generalized (most specific) of the cuboid

→ the Apex cuboid is the most generalized (least specific) of the cuboids

The statement such as

“compute cube sales_cube”

Would explicitly use to compute the sales aggregate cuboids for all 8 subset of the set[city, year, item] including the empty subset

Curse of dimensionality

→ A Major challenge related to precomputation is that the required storage space may explode if all the cuboids in the data cube are precomputed

→ Especially, when cube has many dimensions

Example:

Time is usually explored not only at one conceptual level (e.g. year) but rather at multiple conceptual levels such as in the hierarchy

“day<month<quarter<year”.

For n dimensional data cube, the total number of cuboids that can be generated is:

total no. of cuboids = $\prod_{i=1}^n (L_i + 1)$

where L_i is the number of levels associated with the dimension i including top level all.

1.2 Partial materialization

Materialization is a precomputed table comprising aggregated data from fact and dimension table

There are three choices for data cube materialization:

→ No materialization

→ Full materialization

→ Partial materialization

No materialization:

Do not precompute any of the non-base cuboids.

This leads to computing expensive multidimensional aggregates on the fly, which can be extremely slow.

Full materialization:

Precompute all the cuboids.

The resulting lattice of computed cuboid is referred as “full cube”.

This choice requires huge amount of memory space in order to store all precomputed cuboids

Partial materialization:

Selectively compute a proper subset of the whole set of possible cuboids.

We may compute the subset of the cube, which contains only those cells that satisfy some user-specified criterion.

1.3 Indexing OLAP data

→ Indexing will be done on OLAP data to facilitate efficient data accessing.

→ There are two ways of indexing OLAP data:

Bitmap indexing

Join indexing

Bitmap Indexing

→ Popular indexing method, because it allows quick searching in data cube

The bitmap index is an alternative representation of the record ID (RID) list. In the bitmap index for a given attribute, there is a distinct bit vector, B_v , for each value v in the domain of the attribute. If the domain of a given attribute consists of n values, then n bits are needed for each entry in the bitmap index (i.e., there are n bit vectors). If the attribute has the value v for a given row in the data table, then the bit representing that value is set to 1 in the corresponding row of the bitmap index. All other bits for that row are set to 0.

Bitmap indexing provides reduction in space and Input/Output space

Consider the following example:

Base table

<i>RID</i>	<i>item</i>	<i>city</i>
R1	H	V
R2	C	V
R3	P	V
R4	S	V
R5	H	T
R6	C	T
R7	P	T
R8	S	T

item bitmap index table

<i>RID</i>	H	C	P	S
R1	1	0	0	0
R2	0	1	0	0
R3	0	0	1	0
R4	0	0	0	1
R5	1	0	0	0
R6	0	1	0	0
R7	0	0	1	0
R8	0	0	0	1

city bitmap index table

<i>RID</i>	V	T
R1	1	0
R2	1	0
R3	1	0
R4	1	0
R5	0	1
R6	0	1
R7	0	1
R8	0	1

Note: H for "home entertainment," C for "computer," P for "phone," S for "security," V for "Vancouver," T for "Toronto."

Consider the following example to compute the following query:

Select * from student where gender= M and %above 60= Yes and passport exists= Yes

ROLL NO	NAME	GENDER	% ABOVE 60	PASSPORT EXISTS?
111	Alex	M	Yes	Yes
112	John	M	No	Yes
113	Robert	M	No	Yes
114	Shipra	F	Yes	No
115	Reema	F	No	Yes

Male	Female
1	0
1	0
1	0
0	1
0	1

Yes	No
1	0
0	1
0	1
1	0
0	1

Yes	No
1	0
1	0
1	0
0	1
1	0

ROLL NO	NAME	GENDER	% ABOVE 60	PASSPORT EXISTS?
111	Alex	M	Yes	Yes
112	John	M	No	Yes
113	Robert	M	No	Yes
114	Shipra	F	Yes	No
115	Reema	F	No	Yes

Male	Female
1	0
1	0
1	0
0	1
0	1

Yes	No
1	0
0	1
0	1
1	0
0	1

Yes	No
1	0
1	0
1	0
0	1
1	0

Select * from student where gender= M and %above 60= Yes and passport exists= Yes

ROLL NO	NAME	GENDER	% ABOVE 60	PASSPORT EXISTS?
111	Alex	M	Yes	Yes

Join Indexing

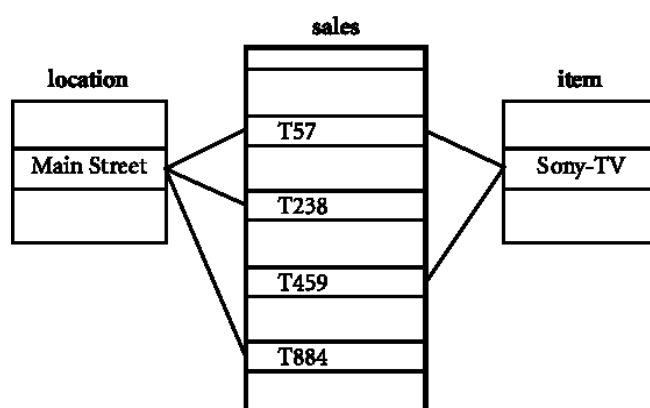
Join indexing registers the joinable rows of two relations from a relational database.

Example:

If two relation R(RID, A) and S(B, SID) join on the attribute A and B, then the join index record contains the pair(RID, SID), where RID and SID are record identifier from the R and S relation.

Join Index records can identify joinable tuples without performing costly join operations.

Consider an example of “AllElectronics” star schema of a join index relationship between the “sales” fact table and the “location” and “item” dimension tables



Above figure shows the linkage between sales fact table and location and item dimension table

**Join index table for
location/sales**

location	sales_key
...	...
Main Street	T57
Main Street	T238
Main Street	T884
...	...

**Join index table for
item/sales**

item	sales_key
...	...
Sony-TV	T57
Sony-TV	T459
...	...

**Join index table linking two dimensions
location/item/sales**

location	item	sales_key
...	...	...
Main Street	Sony-TV	T57
...	...	...

To speed up query processing, the join indexing and bitmap indexing methods can be integrated to form “bitmapped join indices”

1.4 Efficient processing of OLAP Queries

The purpose of materialization cuboid and constructing OLAP index structure is to speed up query processing in data center.

Query processing in data cube should proceed as follows:

→ Determine which operation should be performed on available cuboids

This involves transformation of any selection, projection, roll-up, drill down operation specified in the query into corresponding SQL and OLAP operation.

→ Determine to which materialized cuboids the relevant operation should be performed.

This involves identifying all of the materialized cuboid that may potentially be used to answer the query

Example:

Suppose we create a data cube for AllElectronics
sales_cube[time, item, location]

Dimension hierarchy used are:

For time: day<month<quarter<year

For item: item_name<item_brand<type

For location: street<city<state<country

■ Example

- Let the query to be processed be on $\{brand, province_or_state\}$ with the condition " $year = 2010$ ", and there are 4 materialized cuboids available:

- 1) $\{year, item_name, city\}$
- 2) $\{year, brand, country\}$
- 3) $\{year, brand, province_or_state\}$
- 4) $\{item_name, province_or_state\}$ where $year = 2010$

Which should be selected to process the query?

- Finer-granularity data cannot be generated from coarser-granularity data
- So, cuboid 2 cannot be used as *country* is more general concept than *province_or_state*
- Cuboids 1, 3 and 4 can be used to process the query as
 - they have the same set or superset of the dimensions in the query
 - the selection clause in the query can imply the selection in the cuboid
 - The abstraction levels for the *item* and *location* dimensions in these cuboids are at a finer level than *brand* and *province_or_state*, respectively
 - Cuboid 1 would cost most – *item_name* and *city* at lower level than *brand* and *province_or_state*
 - If few *year* values associated with *items* in the cube, but there are several *item_names* for each *brand*, then cuboid 3 is smaller than cuboid 4

1.5 OLAP server architecture: ROLAP vs MOLAP vs HOLAP

Cubes in data warehouse are stored in three different modes:

Multidimensional OLAP (MOLAP)

Relational OLAP (ROLAP)

Hybrid OLAP (HOLAP)

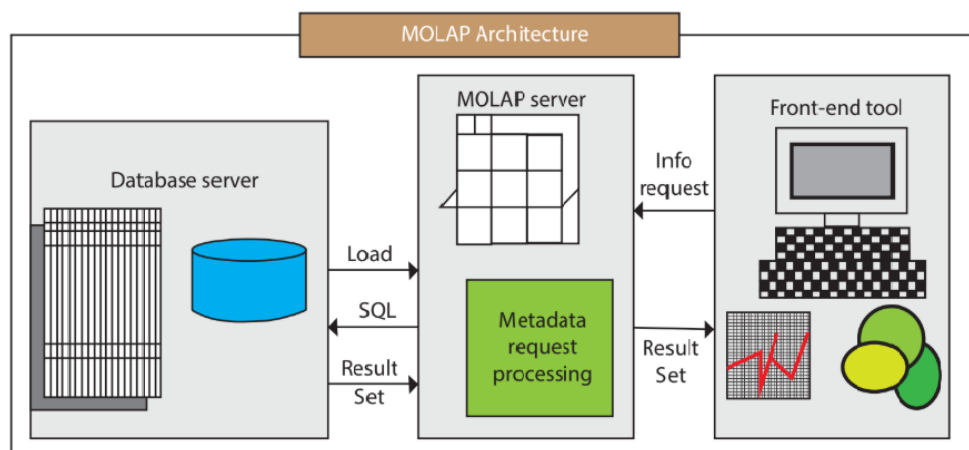
Multidimensional OLAP (MOLAP)

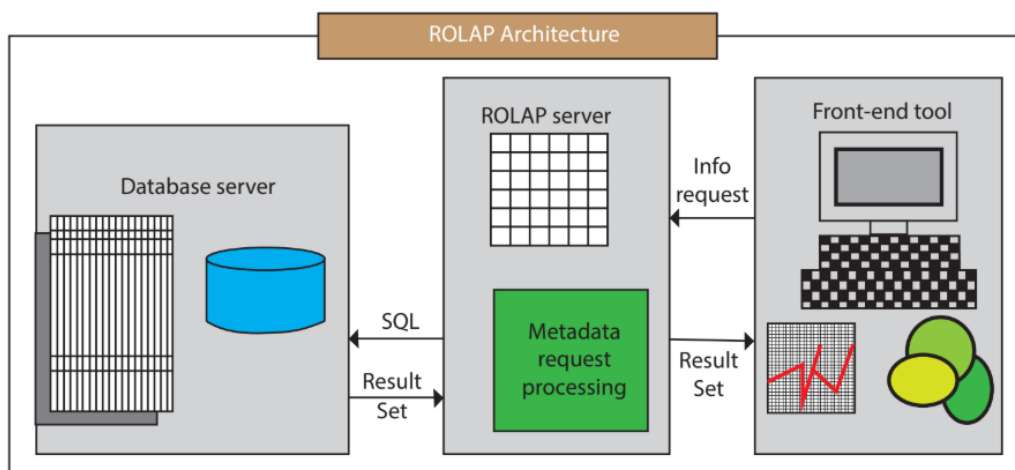
→ Traditional mode in OLAP analysis

→ In MOLAP, data is stored in form of multidimensional data cubes and not in relational database.

→ Advantage in MOLAP is it provides excellent query performance and cube are built for fast data retrieval.

→ Disadvantages of this model are that it can handle only limited amount of data.



Relational OLAP (ROLAP)

- Data in this model is stored in relational data base.
- Advantages of this model is it can handle large volume of data i.e ROLAP is Highly scalable
- In ROLAP, both detailed data and summarized data are stored as shown in figure below.
- Disadvantage is that performance is slow

RID	Item	Day	Month	Quarter	Year	Sales
1001	TV	15	10	Q4	2003	250
1002	TV	23	10	Q4	2003	175
...			...		...	
5001	TV	all	10	Q4	2003	45,786

Hybrid OLAP (HOLAP) servers:

- The hybrid OLAP approach combines ROLAP and MOLAP technology, benefiting from the greater scalability of ROLAP and the faster computation of MOLAP
- HOLAP server may allow large volumes of detail data to be stored in a relational database, while summary data are kept in a separate MOLAP store.

Difference between ROLAP and MOLAP

ROLAP	MOLAP
Stands for Relational Online Analytical Processing	Stands for Multidimensional Online Analytical Processing
Access of ROLAP is slow	Access of MOLAP is fast
Data is stored in relation table	Data is stored in multidimensional array
ROLAP is used for large volume of data	MOLAP is used for limited volume of data
DBMS facility is strong.	DBMS facility is weak.

1.6 Data mining

Data mining is a process of discovering interesting patterns and knowledge from large amount of data.

Data mining as a synonym for term: knowledge discovery from data or KDD (overall process of converting raw data into useful information)

Data mining has also opened up exciting opportunities for exploring and analyzing new types of data and for analyzing different types of data in new ways.

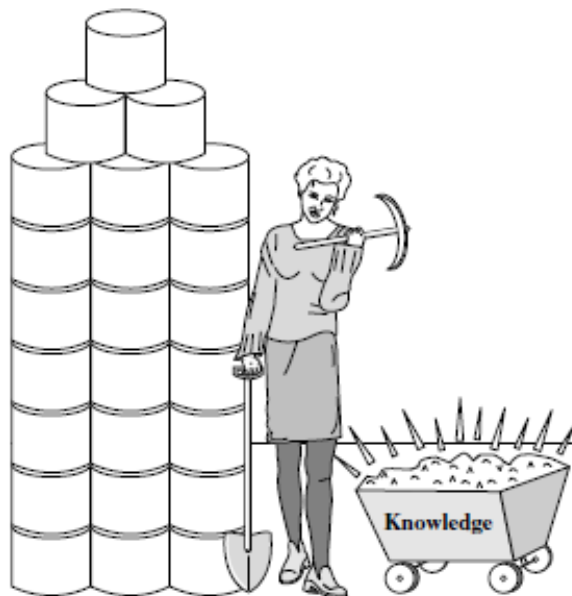
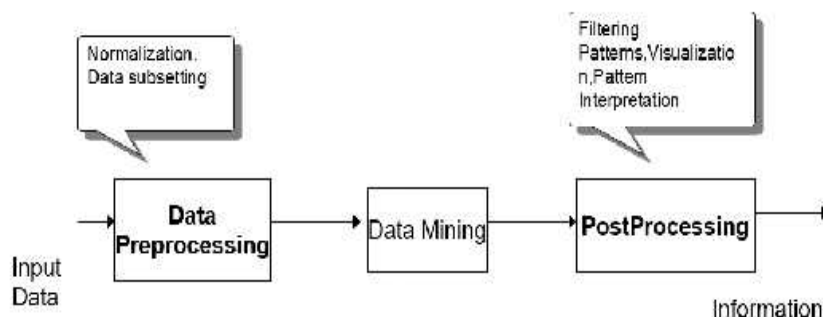


Figure 1.3 Data mining—searching for knowledge (interesting patterns) in data.

1.6 Data Mining and knowledge discovery:

Data mining is an integral part of knowledge discovery in database(KDD), which is an overall process of converting raw data into useful information.

This process consists of series of transitions from “Data Preprocessing” to “post processing” of data mining results.



The input data can be in different formats such as flat files, spread sheets, or relational tables.

The purpose of preprocessing is to transform the raw input data into an appropriate format for subsequent analysis.

Post processing steps ensures that only valid and useful results are incorporated into the decision support system.

KDD Process

The knowledge discovery process is shown in the figure below as an iterative sequence of the following steps:

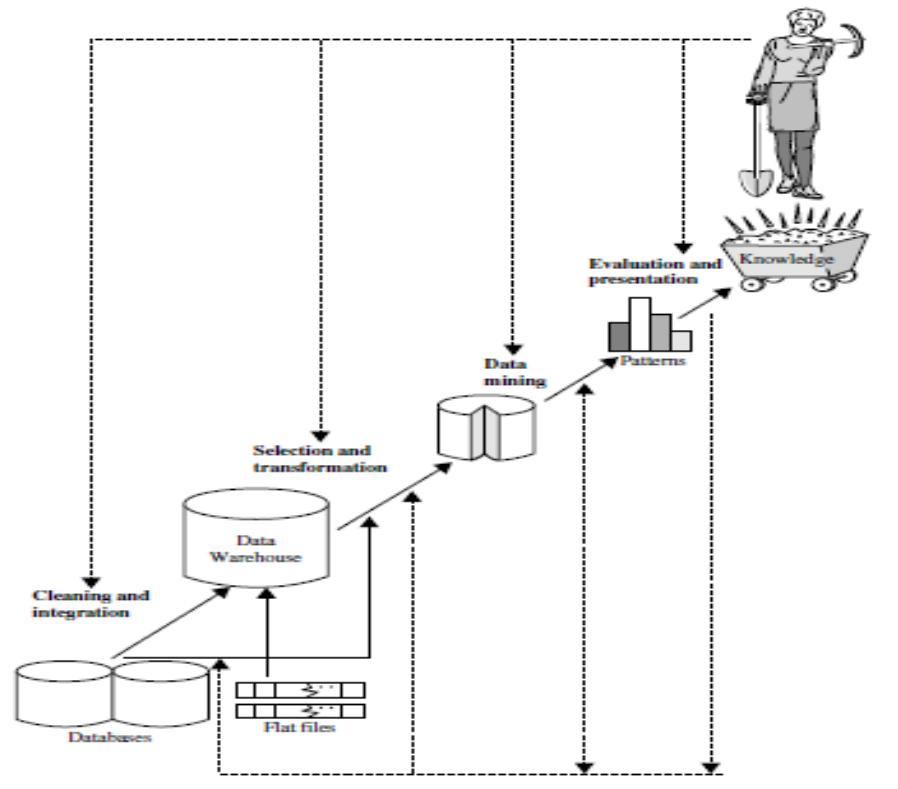


Figure 1.4 Data mining as a step in the process of knowledge discovery.

1. Data cleaning (to remove noise and inconsistent data)
2. Data integration (where multiple data sources may be combined)
3. Data selection (where data relevant to the analysis task are retrieved from the database)
4. Data transformation (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
5. Data mining (an essential process where intelligent methods are applied to extract data patterns)
6. Pattern evaluation (to identify the truly interesting patterns representing knowledge based on interestingness measures)
7. Knowledge presentation (where visualization and knowledge representation techniques are used to present mined knowledge to users)

1.7 Motivating challenges of data mining

- Scalability
- High dimensionality
- Heterogeneous and complex data
- Data ownership and distribution
- Non-traditional analysis

Scalability:

- Because of the advances in data generation and collection, data set with the size of gigabytes, terabytes are becoming common.
- Massive data sets cannot fit into main memory.
- Need to develop scalable data mining algorithms to mine massive data sets.

High dimensionality:

- It is common to encounter data sets with hundreds or thousands of attributes.

- Example: data set that contains measurement of temperature at various location.
- Computational complexity increases as dimensionality increases
- Traditional data analysis techniques were developed for low dimensional data
- Need to develop data mining algorithm to handle high dimensionality

Heterogeneous and complex data:

- Traditional data analysis method deals with data sets containing attributes of same type
- Complex data set contains images, videos, text etc.
- Need to develop mining methods to handle complex data sets.

Data ownership and distribution:

- Data is not stored in one location, or owned by one organization
- Data is geographically distributed among resources belonging to multiple entities
- Need to develop distributed data mining algorithm to handle distributed data sets.

Non-traditional analysis:

- Traditional approach is based on hypothesis and test paradigm
- A hypothesis is proposed, an experiment is designed to gather the data, and then data is analyzed with respect to the hypothesis.
- This process is extremely labor intensive.
- Need to develop data mining methods to automate the process of hypothesis generation and evaluation

1.8 Data Mining Tasks

Data mining tasks are generally divided into two major categories:

- Predictive tasks
- Descriptive tasks

Predictive tasks:

The objective of this task is to predict the values of a particular attribute based on the values of other attributes.

The attribute to be predicted is called: **target or dependent variable**

Attribute used for making prediction are called: **explanatory or independent variable**

Descriptive tasks:

The objective is to derive patterns that summarize the relationship in the data.

Descriptive data mining task require post processing technique to validate and explain the results

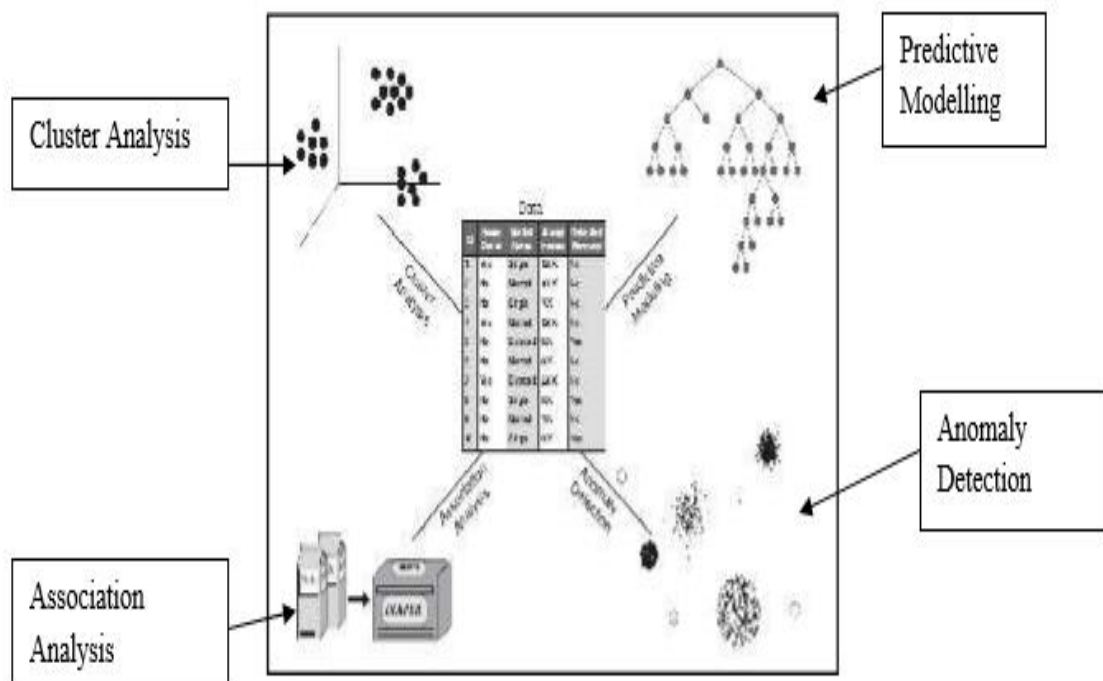


Figure 2.3: Data Mining Tasks

Four of the Core Data Mining tasks:

- 1) Predictive Modelling
- 2) Association analysis
- 3) Cluster analysis
- 4) Anomaly detection

Predictive Modelling

Process that uses data mining and probability to forecast outcomes.

There are two types of predictive modelling tasks:

Classification

Regression

Classification: It is used for discrete target variables

Ex: Web user will make purchase at an online bookstore is a classification task, because the target variable is binary valued.

Regression: It is used for continuous target variables.

Ex: forecasting the future price of a stock is regression task because price is a Continuous values attribute

Example 1(Classification): Predicting the type of a flower

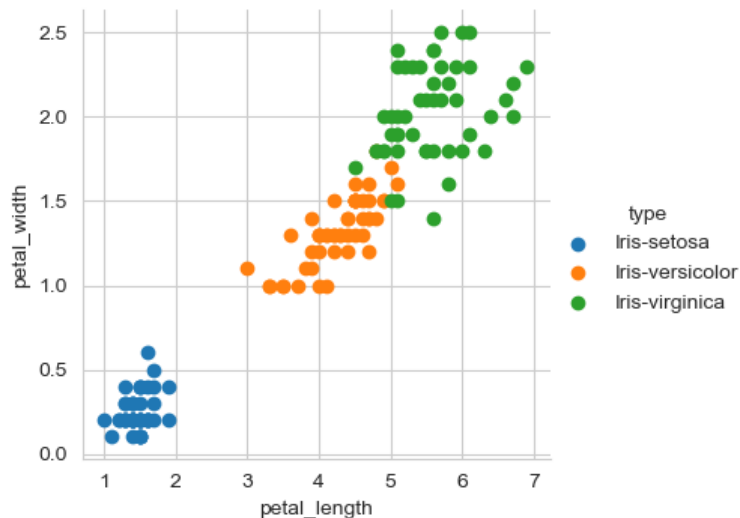
Consider a task of classifying an Iris flower as to whether it belongs to one of the following three Iris species:

Setosa

Versicolor

virginica

To perform this task, we need a data set containing the characteristics of various flowers of these three species.



Above figure shows a plot of petal width versus petal length for 150 flowers in Iris data set. Petal width is broken into the categories low, medium, high. Also petal length is broken into categories low, medium and high.

Based on these categories of petal width and length, following rules can be derived:

Petal width low and Petal length low implies Setosa

Petal width medium and Petal length medium implies Versicolour

Petal width high and Petal high low implies Virginica

Example 2(Regression):

Predicting the income of an Employee based on Education, Experience, and other demographic factors.

Association Analysis

The Goal of association analysis is to extract the most of interesting patterns in an efficient manner

Ex: Market based analysis: We may discover the rule that {diapers} → {Milk}, which suggests that customers who buy diapers also tend to buy milk.

Below table illustrates point-of-sale data collected at the checkout counter of a grocery store.

Transaction ID	Items
1	{Bread, Butter, Diapers, Milk}
2	{coffee, Diapers, Cookies, Milk}
3	{Bread, Butter, Coffee, Diapers, Milk, Eggs}
4	{Bread, Butter, Salmon, Chicken}
5	{Eggs, Bread, Butter}

Cluster Analysis

Used to identify data objects that are similar to one another

Cluster analysis is used to find groups of closely related observations so that observations that belong to the same cluster are more similar to each other than observations that belong to other clusters

Example: Document clustering

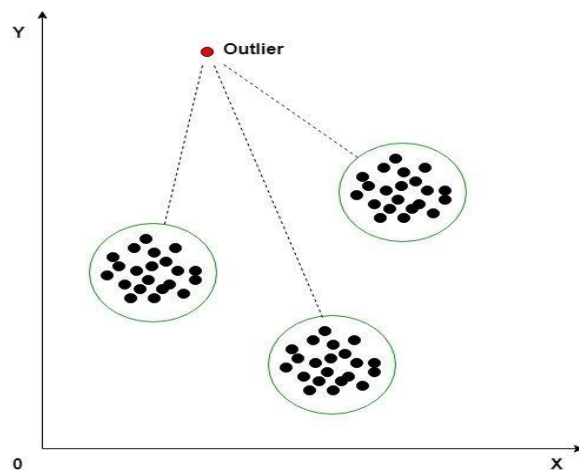
Article	Words
1	Dollar:1, industry:4, country:2, loan:3, government:2
2	Machinery:2, labor:3, market:4, industry:2, work:3, country:1
3	Job:5, inflation:3, rise:2, jobless:2, market: 3, country:2
4	Patient:4, symptoms:2, drug:3, health:2, clinic:2, doctor:2
5	Death:2, cancer:4, drug:3, public:4, health:4, director:1

Collection of news articles shown in the table above can be grouped. Each article is represented as a set of word-frequency pair (w, c) where w is a word and c is the number of times the word appears in the article.

First cluster consists of first three articles, which correspond to the news about the economy while the second cluster contains the last four articles which corresponds to news about health care.

Anomaly Detection

It is the task of identifying observations whose characteristics are significantly different from the rest of the data. Such observations are known as anomalies or outliers.



Applications of anomalies are: fraud detection, network intrusion, unusual patterns of diseases

Example: Credit card fraud detection

A credit card company records transaction made by every credit card holder, along with personal information such as credit card limit, age, annual income, and address.

Since the number of fraudulent case is small compared to the number of legitimate transaction, anomaly detection technique can be applied to build a profile of legitimate transactions for the users. When a new transaction arrives, it is compared against the profile of the user. If the characteristics of the transaction are very different from the previously created profile, then the transaction is flagged as potentially fraudulent.

1.9 Types of Data

What is Data set?

Collection of data objects and their attributes

Other names for a data object are record, point, vector, pattern, event, case, sample, observation.

An attribute is a property or characteristic of an object

Data can be classified into two broad types:

Qualitative

Quantitative

Quantitative data can be counted, measured, and expressed using numbers.

Example: colours of eyes: black, brown, blue

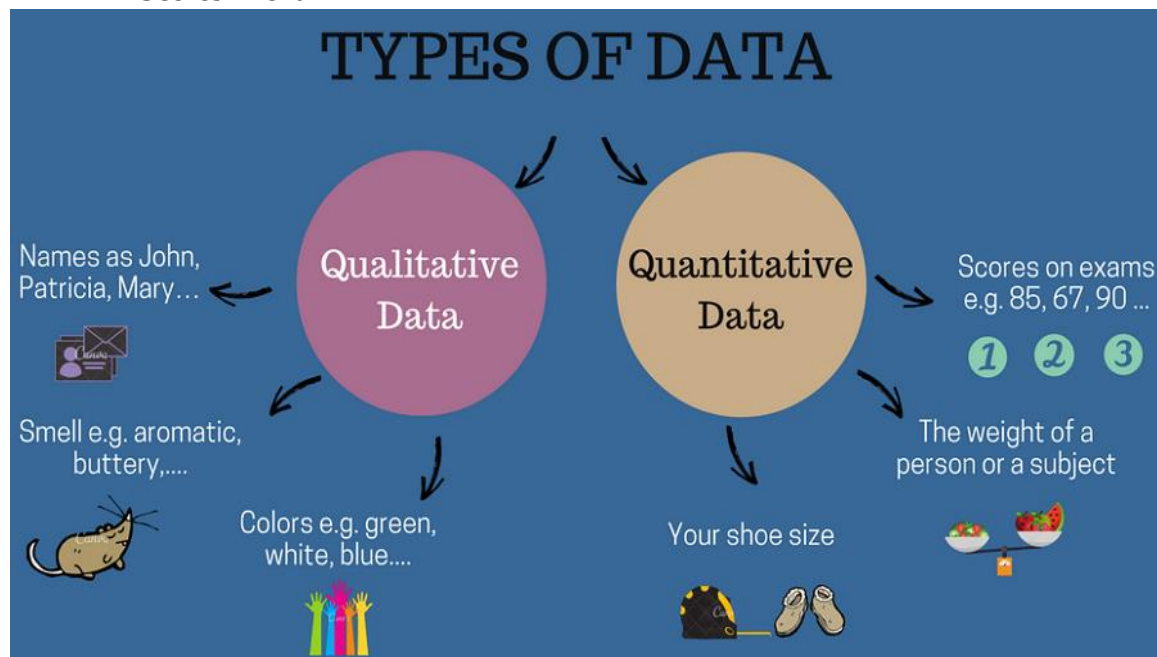
exam result: pass or fail

Qualitative data is descriptive and conceptual.

Example: Height of a person

Shoe size

Scores in exam



1.10 Types of an Attribute

An Attribute is a property or characteristic of an object.

Different types of attributes:

Nominal

Ordinal

Interval

Ratio

Nominal Attributes

Nominal means “relating to names.”

The values of a nominal attribute are just names of things

Nominal values provide only enough information to distinguish one object from another(=, ≠)

Example:

Suppose that hair color and marital status are two attributes

In our application, possible values for hair color are black, brown, blond, red, grey, and white.

The attribute marital status can take on the values single, married, divorced, and widowed. Both hair color and marital status are nominal attributes.

Another example of a nominal attribute is occupation, with the values teacher, dentist, programmer, farmer, and so on.

Ordinal Attributes (<, >, >=, <=)

An ordinal attribute is an attribute with possible values that have a meaningful order or ranking among them

Suppose that drink size corresponds to the size of drinks available at a fast-food restaurant. This attribute has three possible values: small, medium, and large. The values have a meaningful sequence

Other examples of ordinal attributes include grade (e.g., A+, A, A-, B+, and so on)

Professional ranks can be enumerated in a sequential order: for example, assistant, associate, specialist

Customer satisfaction has the following ordinal categories: 0: very dissatisfied, 1: somewhat dissatisfied, 2: neutral, 3: satisfied, and 4: very satisfied.

Interval-Scaled Attributes (+ or -)

Interval-scaled attributes are measured on a scale of equal-size units

Attributes allow us to compare and quantify the difference between values.

For example, a temperature of 20°C is five degrees higher than a temperature of 15°C.

Calendar dates are another example. For instance, the years 2002 and 2010 are eight years apart.

Ratio scaled attributes (* or /)

A measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value

Years of experience (e.g., the objects are employees) and number of words (e.g., the objects are documents).

1.10.1 Describing Attributes by the number of values

Discrete Attribute:

A discrete attribute has a finite set of values.

Example: Marital status (single, married, divorced, widowed)

Binary attribute

Binary attribute is special case of discrete attribute and assume only two values (true/ false, yes/no, male/female, 0/1)

Example: result (pass, fail)

Continuous attribute

Continuous attribute is the one whose values are real numbers

Example: temperature, height or weight

1.11 Characteristics of a Data Sets

Three characteristics that apply to many data sets and have a significant impact on the data mining techniques:

Dimensionality

Sparsity

Resolution

Dimensionality

→ The dimensionality of a data set is the number of attributes that an object in a data possess.

→ Data with the small number of dimensions tends to be qualitatively different than moderate or high dimensional data.

→ Difficulties associated with high dimensional data are sometimes referred to as the curse of dimensionality

→ Example health care data

Sparsity

→ For some data set, most attributes of an object have value 0

→ sparsity is an advantage because usually only the non-zero values need to be stored and manipulated

Resolution

→ It is frequently possible to obtain data at different levels of resolution

- Properties of data are different at different levels of resolution
- Example: The surface of the earth seems very uneven at a resolution of few meters, but is relatively smooth at a resolution of tens of kilometres.
- Pattern of the data depends on the resolution
- If the resolution is too fine, a pattern may not be visible or may be buried in noise.
- If the resolution is too coarse, the pattern may disappear

1.12 Types of data sets

Following are the types of data sets:

- Record data
- Graph based data
- Ordered data

Record data

- Collection of records (data objects), each of which consists of a fixed set of data fields(attributes)
- There is explicit relationship among records or data fields
- Record files are usually stored either in flat files or in relational database
- Types of record data are:

- Transaction data
- Data matrix
- Document term matrix

Transaction data

Transaction data is a special type of record data where each record involves a set of items. Consider the grocery store. The set of products purchased by the customer constitutes a transaction. While individual products that were purchased are the items. This type of data is called market basket data.

Transaction ID	Items
1	{Bread, soda, Milk}
2	{beer, bread}
3	{Beer, soda, Diapers, Milk}
4	{Bread, Butter, Chicken}
5	{soda, Bread, Butter}

Data matrix

- Collection of data with same fixed set of numeric attributes
- These data objects can be interpreted as m by n matrix, where there are m rows, one for each objects and n columns, one for each attribute.
- Also called a pattern matrix

Projection of x Load	Projection of y Load	Distance	Load	Thickness
10.23	5.27	15.22	27	1.2
12.65	6.25	16.22	22	1.1
13.54	7.23	17.34	23	1.2
14.27	8.43	18.45	25	0.9

(c) Data matrix.

Document term matrix

A special case of a data matrix in which the attributes are of the same type and are asymmetric. In this, documents are the rows of this matrix, while the term are the columns

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

(d) Document-term matrix.

Graph based data

→ Convenient and powerful representation of data

→ There are two different types of graph based data

Data with Relationship among objects

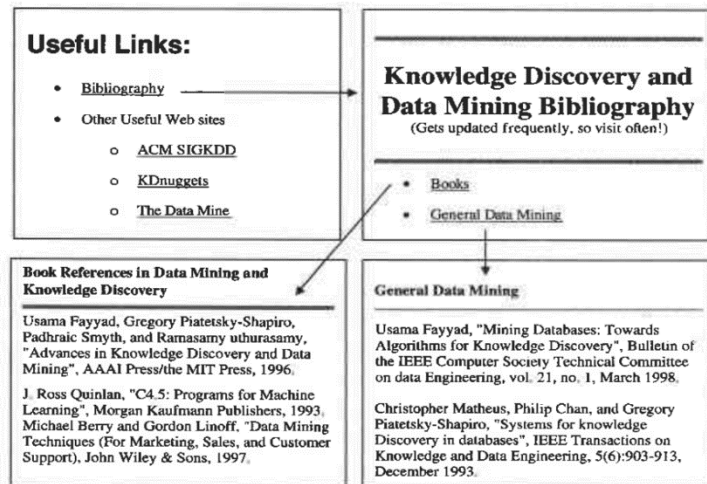
Data with objects that are graphs

Data with Relationship among objects:

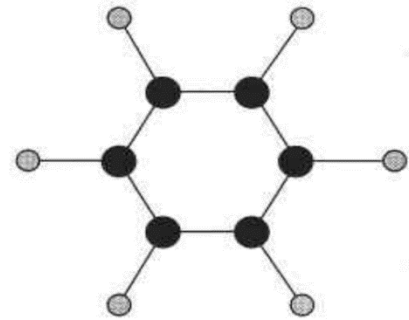
The relationship among the objects frequently convey important information's. In such cases, data is often represented as a graph.

Data objects are mapped to nodes of the graph, while relationship among the objects are mapped to links between objects.

Consider web pages on the world wide web, which contains both text and links to other pages.



(a) Linked Web pages.



(b) Benzene molecule.

Data with objects that are graphs:

If objects have structures, that is objects contain sub-objects that have relationships, then such objects are represented as graphs. For example, the structure of chemical compounds can be represented by a graph where the nodes are atoms and the links between node are chemical bonds. Above figure(b) shows a ball-and-stick diagram of the chemical benzene, which contains atoms of carbon (black) and hydrogen (grey).

Ordered data

→ For some type of data, the attributes have relationship that involve order in time or space.

Ordered data

Different types of ordered data are:

- Sequential data
- Time series data
- Spatial data

Sequential data:

→ Also referred as Temporal data

→ Each record has a time associated with it

→ Consider a retail transaction data set that also stores the time at which the transaction took place.

→ Using this information, it is possible to find patterns such as "people who buy DVD players tends to buy DVDs in the period immediately following the purchases"

Time	Customer	Items Purchased
t1	C1	A, B
t2	C3	A, C
t2	C1	C, D
t3	C2	A, D
t4	C2	E
t5	C1	A, E

Customer	Time and Items Purchased
C1	(t1: A,B) (t2:C,D) (t5:A,E)
C2	(t3: A, D) (t4: E)
C3	(t2: A, C)

(a) Sequential transaction data.

In the above table, there are five different times- t1, t2, t3, t4 and t5

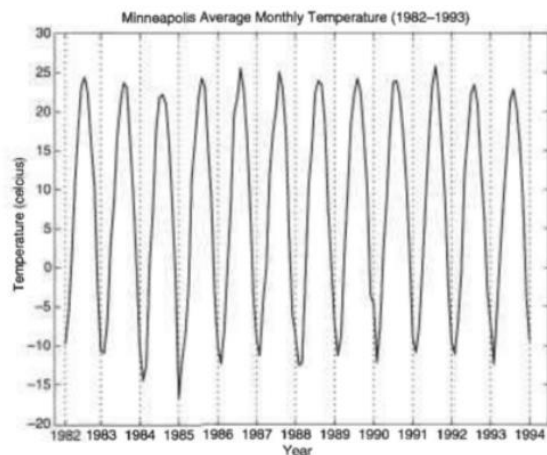
Three different customers- C1, C2, and C3.

Five different items- A, B, C, D and E

In the top table, each row corresponds to the items purchased at a particular time by each customer. For instance, at time t3, customer C2 purchased items A and D. In the bottom table, the same information is displayed, but each row corresponds to a particular customer. Each row contains information on each transaction involving the customer, where a transaction is considered to be a set of items and the time at which those items were purchased. Example: Customer C3 bought items A and C at time t2.

Time series data:

- Time series data is a special type of sequential data in which each record is a time series
- Series of measurements taken over time



(c) Temperature time series.

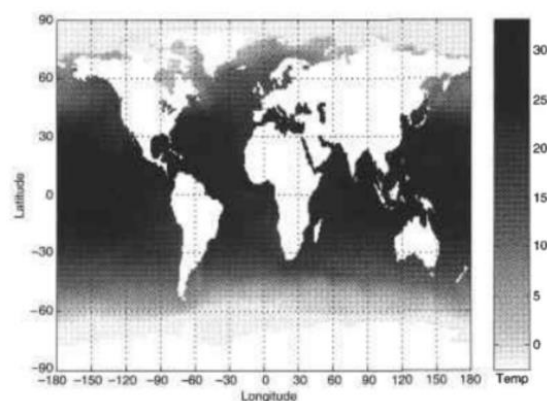
Above figure shows a time series of the average monthly temperature for Minneapolis during the year 1982 to 1994.

If two measurements are close in time, then the values of those measurements are often very similar which is termed as **temporal autocorrelation**.

Spatial Data:

- Some objects have spatial attributes such as positions and areas
- Example: Weather data

Objects that are physically close usually have similar values. For instance, two points on earth that are close to each other have similar values for temperature and rainfall.



(d) Spatial temperature data.

1.13 Data Quality

- Today's real world databases are highly susceptible to noise and inconsistent data due to their huge size

→ Data mining applications focuses on:

- * Detection and correction of data Quality problems (data cleaning)
- * Use of algorithms that can tolerate poor data quality

Noise and artifacts

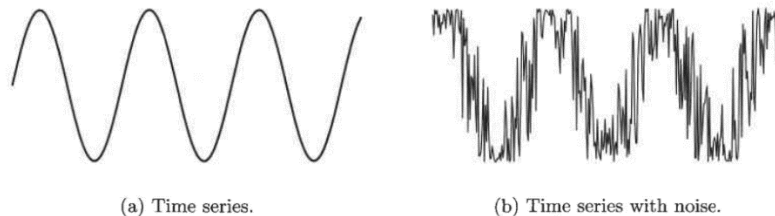
Noise is a random error or variance in a measured variable.

→ It may involve the distortion of values or addition of spurious objects

Distortion of the data are referred as **artifacts**

Below figure shows a time series before and after it has been disrupted by random noise.

If more noise were added to the time series, its shape would be lost.



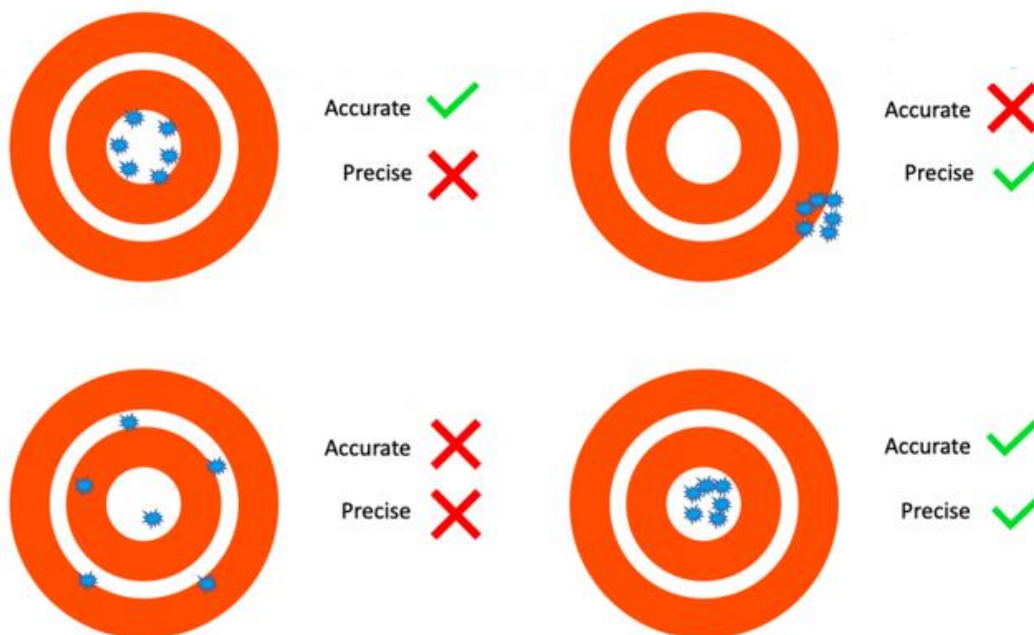
Precision, Bias and Accuracy

Precision: The closeness of repeated measurements to one another

Bias: Measure of how far the expected value is from the true value

Accurate: Closeness of the measurement to the true value of the quantity being measured.

Consider the following Example:



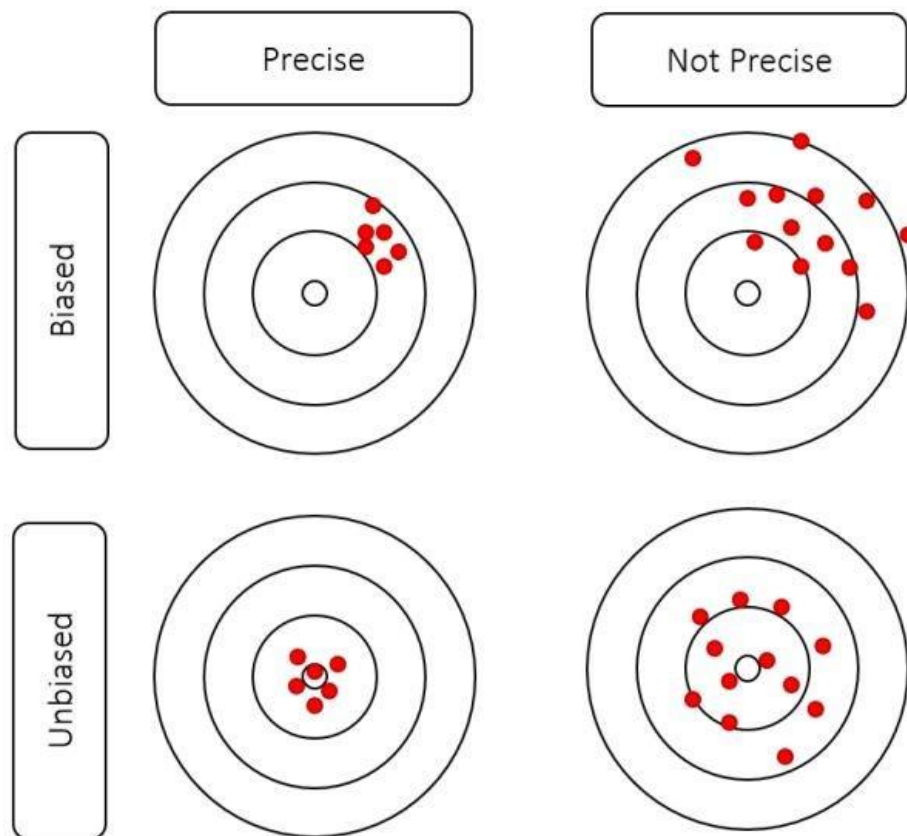


Figure 2.1: Precision & Bias

Outlier

Outliers are data objects that have characteristics that are different from most of the other data objects in the data set

- An outlier can cause serious problem in analysis
- Unlike noise, outliers may sometimes be of interest
- For example, in fraud and network intrusion detection, the goal is to find unusual objects from large number of normal ones

Missing values

- Not unusual for an object to be missing one or more attribute values
- In some cases, information is not collected
- Example: Some people decline to give their age or weight

Missing values should be taken into account during the data analysis.

There are several strategies for dealing with missing data.

Imagine that you need to analyse AllElectronics sales and customer data.

You note that many tuples have not recorded value for several attributes, such as customer income.

How can we fill in the missing values for this attribute?

1. Ignore the tuple or Attribute:

This method is not very effective, unless the tuple contains several attributes with missing values.

2. Fill in the missing value manually:

This approach is time consuming and may not be feasible given a large data set with many missing values.

3. Use a global constant to fill in the missing value:

Replace all missing attribute values by the same constant such as a label like “Unknown”.

4. Use a measure of central tendency for the attribute (e.g., the mean or median) to fill in the missing value:

For example, suppose that the average income of AllElectronics customers is \$56,000. Use this value to replace the missing value for income.

5. Use the most probable value to fill in the missing value:

Using the other customer attributes in your data set, you may construct a decision tree to predict the missing values for *income*.

Inconsistent values

→ Data can contain inconsistent values

→ Example: Consider an address field where both a zip code and city are listed, but the specified zip code area is not contained in that city

→ Some type of inconsistencies are easy to detect

→ Example: Person's height cannot be negative

→ Once an inconsistency has been detected, it is sometimes possible to correct the data

Duplicate data

→ A data set may include data objects that are duplicates or almost duplicate of one another

→ To detect and eliminate such duplicates two main issues must be addressed:

* If there are two objects that actually represent a single object then the values of corresponding attributes may differ, these inconsistent values must be resolved.

* Care needs to be taken to avoid accidentally combining data objects that are similar, but not duplicates such as two distinct people with identical names

The term **deduplication** is often used to refer to the process of dealing with these issues

1.14 Data Preprocessing

Data Preprocessing is a Data mining technique that involves transforming raw data as understandable format

→ Real world data is often incomplete, inconsistent and likely to contain many errors

→ Data preprocessing resolves such issues

Important Approaches for data Pre Processing are:

Aggregation

Sampling

Dimensionality reduction

Feature subset selection

Feature creation

Discretization and binarization

Variable transformation

Aggregation

→ "less is more"

→ Aggregation is combining two or more objects into single object.

→ Disadvantages of aggregation is the potential loss of interesting details

→ Advantage: requires less memory, and processing time

Table 2.4. Data set containing information about customer purchases.

Transaction ID	Item	Store Location	Date	Price	...
⋮	⋮	⋮	⋮	⋮	⋮
101123	Watch	Chicago	09/06/04	\$25.99	...
101123	Battery	Chicago	09/06/04	\$5.99	...
101124	Shoes	Minneapolis	09/06/04	\$75.00	...
⋮	⋮	⋮	⋮	⋮	⋮

Consider the above transaction, recording the daily sales of product in various store locations. For different days over the course of a year.

One way to aggregate transaction for this data set is to replace all the transaction of a single store with a single storewide transaction.

This reduces the hundreds or thousands of transactions that occurs daily at a specific store to a single daily transaction.

Quantitative attributes such as price, are typically aggregated by taking sum or an average.

A Qualitative attribute such as item can either be omitted or summarized as the set of all the items that were sold at that location.

Year 2004	
Quarter	Sales
Q1	\$386,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year 2003	
Quarter	Sales
Q1	\$386,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year 2002	
Quarter	Sales
Q1	\$224,000
Q2	\$408,000
Q3	\$350,000
Q4	\$586,000

Year	Sales
2002	\$1,568,000
2003	\$2,356,000
2004	\$3,594,000

Sales data for a given branch of *AllElectronics* for the years 2002 to 2004. On the left, the sales are shown per quarter. On the right, the data are aggregated to provide the annual sales.

Sampling

Sampling can be used as a data reduction technique because it allows a large data set to be represented by a much smaller random data sample (or subset).

- Approach used for selecting a subset of the data objects to be analyzed
- Sampling can be very useful in data mining.

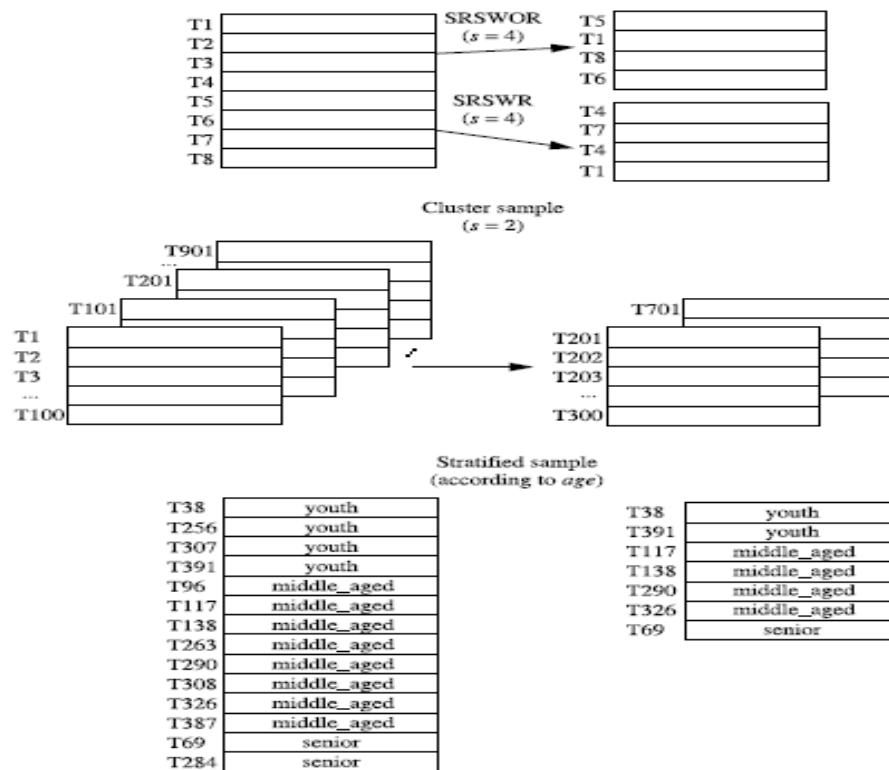
Sampling Approaches

- Simple random sample without replacement (SRSWOR)
- Simple random sample with replacement (SRSWR)
- Cluster sample
- Stratified sample

There is an equal probability of selecting any particular item

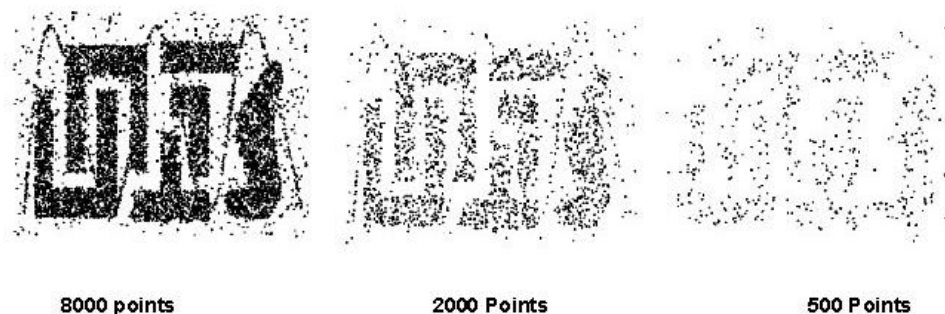
There are two variations on random sampling:

- 1) Sampling without replacement: As each item is selected, it is removed from the population
- 2) Sampling with replacement: Objects are not removed from the population as they are selected for the sample. In sampling with replacement, the same object can be picked up more than once
- 4) Cluster sample: Random documents are removed from the set of documents
- 3) Stratified sampling – Equal numbers of objects are drawn from each group though the groups are different size



Once a sampling technique has been selected, it is still necessary to choose the sample size. Larger sample size increase the probability that a sample will be representative, but they also eliminate much of the advantage of the sampling.

With smaller sample sizes, pattern may be missed or erroneous pattern can be detected.



Progressive sampling:

- Proper sampling size is difficult to determine
- So adaptive or progressive sampling schemes are sometimes used
- Start with a small sample and then increase the sample size until the sample of sufficient size has been obtained.
- This technique eliminates the need to determine the correct sample size initially.

Dimensionality Reduction

Data analysis becomes harder as the dimensionality of the data increases.

The term dimensionality reduction is often reserved for those techniques that reduce the dimensionality of a data set by creating new attributes that are a combination of old attributes.

Purpose:

- o Avoid curse of dimensionality
- o Reduce amount of time and memory required by data mining algorithms
- o Allow data to be more easily visualized
- o May help to eliminate irrelevant features or reduce noise

Curse of Dimensionality: The curse of dimensionality refers to various phenomena that arise when analysing and organizing data in high-dimensional spaces (often with hundreds or thousands of dimensions) that do not occur in low-dimensional settings such as the three-dimensional physical space of everyday experience.

Amount of data we need: Obviously when numbers of dimensions are more we need a bigger data set. Because only with large dataset we can get more information in higher dimensions. Therefore, as number of dimensions' increases amount of data we need to generalize accuracy grows exponentially.

Feature subset selection

Another way to reduce Dimensionality of data and this is not applicable for redundant and irrelevant features.

Redundant Feature

Duplicate information contained in one or more other attributes

Example: purchase price of product and sales tax paid contains same information

Irrelevant Feature

Contains information that is not useful for data mining task

Example: Student ID is irrelevant to the task of predicting their marks

Three standard approaches to feature selection:

Embedded approach

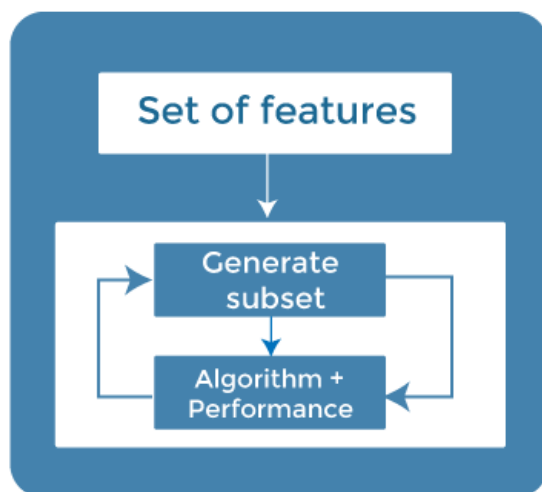
Filter approach

Wrapper approach

Embedded Approach

→ Feature selection occurs naturally as part of the data mining algorithm.

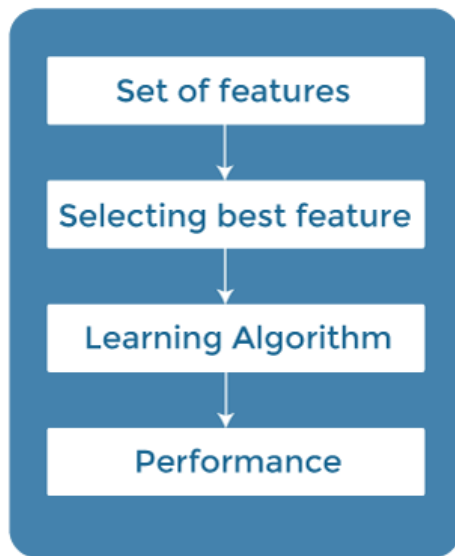
→ Algorithm itself decides which attribute to use and which to ignore



Filter Approaches

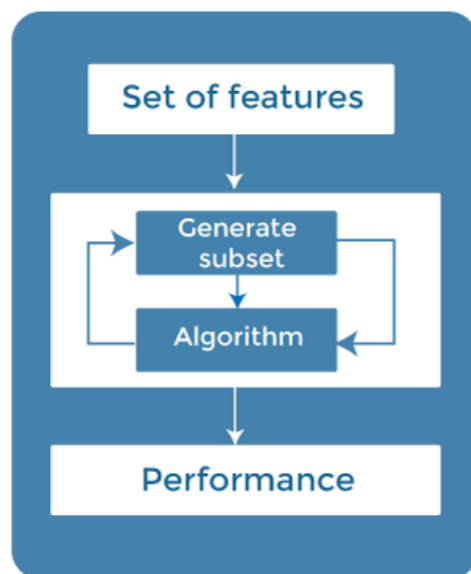
→ Features are selected before data mining algorithm runs

→ Approach independent of data mining task



Wrapper Approach

→ Method uses data mining algorithm as a black box to find best subset of attributes



An Architecture for feature subset selection



Flowchart of a feature subset selection process.

The feature selection process is viewed as consisting of four parts:

A measure for evaluating subset, a search strategy that controls the generation of a new subset of features, a stopping criteria and a validation procedure.

→ feature subset selection is a search over all possible subsets of features.

Many different types of search strategies can be used, but the search should be computationally inexpensive.

Feature Weighting:

→ Alternative to eliminate features

→ More important features are assigned a higher weight, while less important feature are given lower weight

→ Weights are assigned based on the knowledge about the importance of features

→ Features with larger weights play a more important role in the model.

Feature Creation

→ From the original attributes, a new set of attributes can be created

→ Number of new attributes can be smaller than the number of original attributes

→ Such attributes can be generated using three methodologies:

- * Feature extraction
- * Mapping data to a new space
- * Feature construction

Feature Extraction:

→ The creation of new set of features from the original raw data is known as feature extraction

→ These new reduced set of features should then be able to summarize most of the information contained in the original set of features.

Example: extracting edges from images

Mapping data to a new space:

→ Sometimes a totally different view of data can reveal important and interesting features

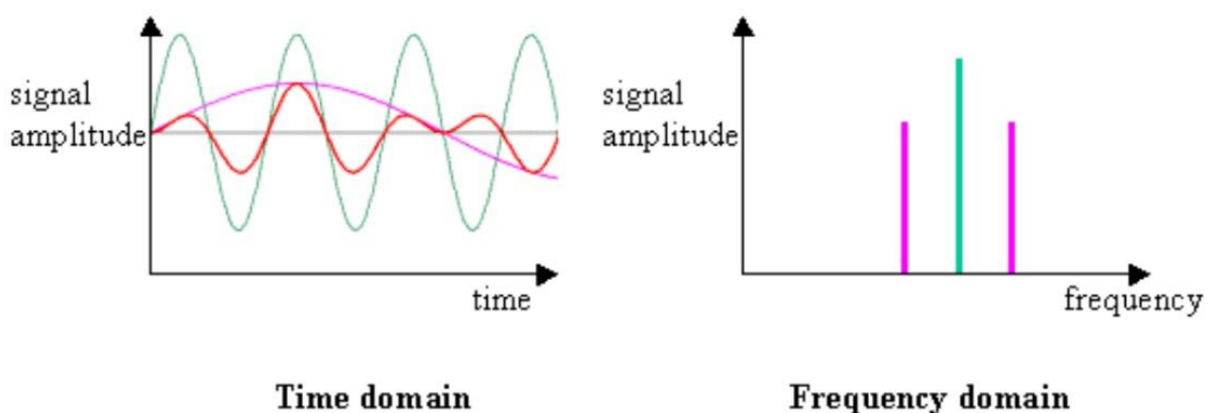
Example:

Consider, for example, time series data, which often contains periodic patterns.

If there is only a single periodic pattern and not much noise then the pattern is easily detected.

If, on the other hand, there are a number of periodic patterns and a significant amount of noise is present, then these patterns are hard to detect.

Such patterns can be detected by applying a Fourier transform to the time series in order to change to a representation in which frequency information is explicit



Feature construction:

→ Sometimes the features in the original data sets have the necessary information, but it is not in a form suitable for the data mining algorithm.

→ In this situation, one or more new features constructed out of the original features can be more useful than the original features.

→ Example: dividing mass by volume to get density

Discretization and Binarization

→ **Discretization** is the process of converting a continuous attribute into an ordinal attribute.

→ A potentially infinite number of values are mapped into a small number of categories.

→ Example:

height measured as {low, medium, high}

age measured as {youth, middle-aged, senior}

→ **Binarization** maps a continuous or categorical attribute into one or more binary variables

→ Often convert a continuous attribute to a categorical attribute and then convert a categorical attribute to a set of binary attributes

Conversion of a categorical attribute to three binary attributes

Categorical Value	Integer Value	x_1	x_2	x_3
<i>awful</i>	0	0	0	0
<i>poor</i>	1	0	0	1
<i>OK</i>	2	0	1	0
<i>good</i>	3	0	1	1
<i>great</i>	4	1	0	0

Conversion of a categorical attribute to five asymmetric binary attributes

Categorical Value	Integer Value	x_1	x_2	x_3	x_4	x_5
<i>awful</i>	0	1	0	0	0	0
<i>poor</i>	1	0	1	0	0	0
<i>OK</i>	2	0	0	1	0	0
<i>good</i>	3	0	0	0	1	0
<i>great</i>	4	0	0	0	0	1

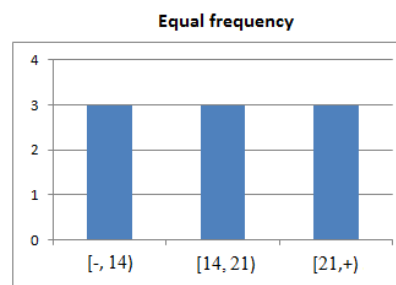
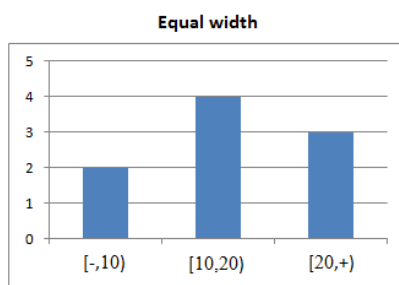
Unsupervised discretization methods:

Equal-width

Equal-frequency

Equal-width approach divides the range of the attributes into a user specified number of intervals each having the same width

Equal-frequency puts the same number of objects into each interval



Consider the following example for Equal Width Interval Discretization:

$$\text{Interval} = \frac{V_{\max} - V_{\min}}{k}$$

$$\text{Boundaries} = V_{\min} + (i \times \text{Interval})$$

Table 1. Solved example for Equal Width Discretization

Data values	V1	V2	V3	V4	V5	V6	V7	V8
Original Continuous Values	10	50	15	20	12	25	40	30
Sorted values	10	12	15	20	25	30	40	50
Description	Here $V_{\min}=10$ $V_{\max}=50$ let $k=2$ So Interval=20 and Boundary=10+20=30							
Discrete Values	1	2	1	1	1	1	2	1

Consider the following example for Equal-Frequency Interval Discretization:

Table 2. Solved example for Equal Frequency Discretization

Data values	V1	V2	V3	V4	V5	V6	V7	V8
Original Continuous Values	10	50	15	20	12	25	40	30
Sorted values	10	12	15	20	25	30	40	50
Description	Let $k=2$ So each interval will contain $8/2=4$ data values							
Discrete Values	1	2	1	1	1	2	2	2

Variable transformation

An attribute transform is a function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values

Two types of variable transformation:

Simple functional transformation

Normalization or Standardization

Simple functional transformation is applied to each value individually

If x is a variable, then Simple Functions include:

$$e^x$$

$$x^k$$

$$\log x$$

$$1/x$$

$$\sqrt{x}$$

$$\sin x$$

The goal of Normalization or Standardization is to make an entire set of values have a particular property

Min-max normalization:

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A.$$

Min-max normalization:

Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000, respectively.

We would like to map *income* to the range [0.0,1.0].

By min-max normalization, a value of \$73,600 for *income* is transformed to

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716.$$

z-score normalization (zero-mean normalization):

$$v' = \frac{v - \bar{A}}{\sigma_A},$$

Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for *income* is transformed to

$$\frac{73,600 - 54,000}{16,000} = 1.225.$$

1.15 Measures of Similarity and dissimilarity

Term **proximity** is used to refer to either similarity or dissimilarity between the objects

Similarity

Similarity between two objects is a numerical measure of the degree to which the two objects are alike

Similarities are usually non-negative and are often between 0(no similarity) or 1(complete similarity)

Dissimilarity

Dissimilarity between two objects is a numerical measure of the degree to which the two objects are different

Term **distance** is used as a synonym for dissimilarity

Dissimilarity sometimes fall in the interval [0,1], but it is also common for them to range from 0 to ∞

Similarities and dissimilarities for simple attributes

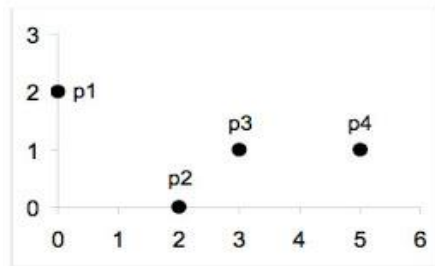
Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = x - y / (n - 1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = x - y $	$s = -d, s = \frac{1}{1+d}, s = e^{-d},$ $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

Dissimilarities between data objects

Euclidean distance:

Euclidean distance, d , between two points x and y , in one-, two-, three, or higher dimensional space is:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

Euclidean distance equation is generalized by Minkowski distance:

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r},$$

r is a parameter

Common examples of Minkowski distances:

r=1 City block/Manhattan, taxicab, L1 norm

r=2 Euclidean distance, L2 norm

r=∞ Supremum, Lmax or L∞ norm(maximum distance between any two attributes)

$$d(\mathbf{x}, \mathbf{y}) = \lim_{r \rightarrow \infty} \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}.$$

Properties of Euclidean distance

If d(x,y) is the distance between two points, x and y, then the following properties hold:

1. Positivity:

- a) $d(x,y) \geq 0$ for all x and y
- b) $d(x,y) = 0$ only if $x=y$

2. Symmetry:

$$d(x,y) = d(y,x) \text{ for all } x \text{ and } y$$

3. Triangle Inequality:

$$d(x,z) \leq d(x,y) + d(y,z) \text{ for all points } x,y,z$$

Similarity measure for binary data

Similarity measure between objects that contain only binary attributes are called similarity coefficients.

Let x and y be two objects that consists of n binary attributes

The comparison of two binary vector leads to the four quantities or frequencies:

f_{00} = the no of attributes where x is 0 and y is 0

f_{01} = the no of attributes where x is 0 and y is 1

f_{10} = the no of attributes where x is 1 and y is 0

f_{11} = the no of attributes where x is 1 and y is 1

Simple Matching Coefficient(SMC)

SMC is defined as:

$$SMC = \frac{\text{number of matching attribute values}}{\text{number of attributes}} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}.$$

Jaccard Coefficient

Suppose x and y are data objects that represent two rows of transaction matrix, then, If asymmetric binary attribute corresponds to an item in a store, then a 1 indicates that the item was purchased, while 0 indicates that the product was not purchased.

Jaccard Coefficient is symbolized by J, is given the following equation:

$$J = \frac{\text{No of Matching presence}}{\text{no of attributes not involved in 00 matches}} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

Calculate the SMC and J for the following two binary vectors:

$$x = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$$

$$y = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)$$

$$f_{00} = 7 \quad \text{the no of attributes where x is 0 and y is 0}$$

$$f_{01} = 2 \quad \text{the no of attributes where x is 0 and y is 1}$$

$$f_{10} = 1 \quad \text{the no of attributes where x is 1 and y is 0}$$

$$f_{11} = 0 \quad \text{the no of attributes where x is 1 and y is 1}$$

$$\text{SMC} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{0 + 7}{2 + 1 + 0 + 7} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{0}{2 + 1 + 0} = 0$$

Cosine Similarity

Documents are often represented as vectors, where each attribute represents the frequency with which a particular term(word) occurs in the document.

Cosine Similarity is one of the most common measure of document similarity

If x and y are two document vectors, then

$$\frac{x \cdot y}{||x|| \ ||y||}$$

. indicates vector dot product

$$x \cdot y = \sum_{k=1}^n x_k y_k$$

$||x||$ is the length of vector x

$$||x|| = \sqrt{\sum_{k=1}^n x_k^2} = \sqrt{x \cdot x}$$

Calculate the cosine similarity for the following two data objects,

$$x = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$$

$$y = (1, 0, 0, 0, 0, 0, 1, 0, 2)$$

$$x \cdot y = 3 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + 5 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 0 + 0 \cdot 2 = 5$$

$$||x|| = \sqrt{3 \cdot 3 + 2 \cdot 2 + 0 \cdot 0 + 5 \cdot 5 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 0 \cdot 0} = 6.48$$

$$||y|| = \sqrt{1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 2 \cdot 2} = 2.24$$

$$\cos(x, y) = 0.31$$